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Wheel bearing

Abstract

A bearing assembly includes an inner bearing ring, an outer bearing ring and rolling bodies located between the inner bearing ring and the outer bearing ring. A gap is defined by a radially outer surface of the inner bearing ring and a radially inner surface of the outer bearing ring. The radially inner surface of the outer bearing ring includes a first radial surface portion that is offset a greater distance from a center of the bearing than a second radial surface portion. The first and second radial surface portions are connected by a sloped surface. A sealing arrangement located in the gap. The sealing arrangement includes a first annular body including an elastomeric sealing protrusion to contact the sloped surface of the outer bearing ring.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/892,204 filed on Aug. 22, 2022, which is incorporated by reference herein in its entirety.

FIELD OF INVENTION

(1) The present disclosure relates generally to bearings, and more particularly to bearings with a sealing arrangement.

BACKGROUND

(2) In typical automotive wheels, it is important to protect the rolling elements of the bearings from dust and debris using annular seals. The seals are inserted between the inner and outer bearing rings. To achieve the desired seal performance, normally one or more dynamic lip seals that are located on a mounting body connected to one of the bearing rings are in contact with opposite rotating surface(s) of the other bearing ring at one or more positions. Typical seal designs also provide a sealed connection between the mounting body and the associated bearing ring using a static rubber seal that contacts a part of the bearing ring where the seal mounting body is pressed into position. However, this static seal has a high likelihood of shearing off during assembly of the seal. Additionally, a position of the seal mounting body when installed is also critical to having the correct (i.e. optimal) contact pressure of the lip seal(s).

(3) Conventional seals and sealing components have generally been considered satisfactory for their intended purpose. However, there is still a need in the art for improved bearing seals that provide improved protection, and better installation technique. The present disclosure provides a solution for this need.

SUMMARY

(4) A bearing assembly for a bearing is disclosed. The bearing assembly includes an inner bearing ring, an outer bearing ring, rolling bodies located between the inner bearing ring and the outer bearing ring. The bearing assembly also includes a gap defined by a radially outer surface of the inner bearing ring and a radially inner surface of the outer bearing ring, the radially inner surface of the outer bearing ring includes a first radial surface portion that is offset a greater distance from a center of the bearing assembly than a second radial surface portion, the first and second radial surface portions are connected by a sloped surface. A sealing arrangement located can be in the gap, which includes, a first annular body to contact the sloped surface of the outer bearing ring, and an axially extending first leg configured for a press fit with the first radial surface portion in an installation position.

(5) In further aspects, the axially extending first leg can include a radially inwardly directed bend that transitions to a radially extending second leg, and an elastomeric sealing protrusion on the first annular body, adjacent to the sloped surface, can be located in an area of the radially inwardly directed bend. The first annular body can include an axially extending third leg, and a first elastomeric lip seal that can contacts the inner bearing ring. The bearing assembly can include a second annular body having an axially extending fourth leg press fit to the radially outer surface of the inner bearing ring, wherein the fourth axially extending leg can transitions into a radially extending fifth leg, and an axially extending sixth leg extending from a radially outer end of the fifth leg can define a generally U-shaped cross-section. The third leg can include a radially outwardly directed bend, the third leg can extend into the generally U-shaped cross-section to define a labyrinth seal. A second elastomeric lip seal can extend from the first annular body to the second annular body.

(6) In further aspects, the bearing assembly can include an elastomeric sealing protrusion on the first annular body adjacent to the sloped surface, wherein the elastomeric sealing protrusion can include a maximal outer diameter greater than a diameter defined by the second radial surface portion of the outer bearing ring. The elastomeric sealing protrusion can include a maximal outer diameter smaller than an outer diameter of the first leg.

(7) In further aspects, the first annular body can be metallic. The first annular body can be non-magnetic. The second annular body can include an encoder. The encoder can be located on an axially exterior side of second annular body. The encoder can be magnetized on an axially exterior side thereof or on a radially inward side thereof. The first annular body and the encoder can define at least one gap therebetween.

(8) A method of inserting a bearing sealing arrangement is also disclosed. The method includes inserting a first annular body including in a gap between an inner bearing ring and an outer bearing ring, contacting a radially outer surface of the inner bearing ring with a first elastomeric lip seal that extends from the first annular body, contacting a first radial surface portion of a radially inner surface of the outer bearing ring with a first leg of the first annular body. Then contacting a sloped surface, adjacent to the first radial surface portion of the radially inner surface of the outer bearing ring, with the first annular body and contacting the radially outer surface of the inner bearing ring with a second annular body of the sealing arrangement.

(9) In further aspects, the first annular body can include an elastomeric sealing protrusion, which is inserted in the gap before the second annular body. The second radial surface portion can be located radially inward of the first surface portion. The first leg of the first annular body can be press fit against the sloped surface. The first leg can include a radially inwardly directed bend that transitions to a radially extending second leg, and the elastomeric sealing protrusion can be located in an area of the radially inwardly directed bend.

(10) It is noted that one or more of the optional features noted above may be used alone or in various combinations with one another in connection with the basic bearing assembly and associated sealing arrangement.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that those skilled in the art to which the subject disclosure appertains will readily understand how to make and use the devices and methods of the subject disclosure without undue experimentation, embodiments thereof will be described in detail herein below with reference to certain figures, wherein:

(2) FIG. 1 is a cross-sectional view illustrating a bearing assembly according to one embodiment the disclosure;

(3) FIGS. 2A-2C are cross-sectional views of the bearing assembly of FIG. 1 showing a step-by-step insertion of the sealing arrangement into a gap between inner and outer bearing rings;

(4) FIG. 3 is a cross-sectional view illustrating another embodiment of a bearing assembly according to the disclosure; and

(5) FIG. 4 is a cross-sectional view illustrating another embodiment of a bearing sealing assembly according to the disclosure.

DETAILED DESCRIPTION

(6) Certain terminology is used in the following description for convenience only and is not limiting. The words “inwardly” and “outwardly” refer to directions toward and away from the parts referenced in the drawings. “Axially” refers to a direction along the axis of a shaft. “Radially” refers to a direction normal to an axis. A reference to a list of items that are cited as, for example, “at least one of a or b” (where a and b represent the items being listed) means any single one of the items a or b, or a combination of a and b. This would also apply to lists of three or more items in like manner so that individual ones of the items or combinations thereof are included. The terms “about” and “approximately” encompass + or -10% of an indicated value unless otherwise noted. The terminology includes the words specifically noted above, derivatives thereof and words of similar import.

(7) Reference will now be made to the drawings wherein like reference numerals identify similar structural features or aspects of the subject disclosure. For purposes of explanation and illustration, and not limitation, a system view of an exemplary embodiment of the bearing assembly in accordance with the disclosure is shown in FIG. 1 and is designated generally by reference character **100**. Other embodiments of the assembly in accordance with the disclosure, or aspects thereof, are provided in FIGS. 2A-4, as will be described. The assembly described herein can be used to improve the useful life of the sealing arrangement by improving installation results.

(8) FIG. 1 shows a cross-section of a bearing assembly **100** of a vehicle wheel. The bearing assembly **100** includes an inner bearing ring **102** and an outer bearing ring **104** arranged with rolling bodies **106** therebetween. A gap **108** between the inner bearing ring **102** and the outer bearing ring **104** provides space for a sealing arrangement **110**. The gap **108** is defined by a radially outer surface **112** of the inner bearing ring **102** and a radially inner surface **114** of the outer bearing ring **104**. The radially inner surface **114** of the outer bearing ring **104** can include at least a first radial surface portion **116** and a second radial surface portion **118**. The second radial surface portion **118** is radially inward of the first radial surface portion **116** and the two radial surface portions **116**, **118** are connected by a sloped surface **124**, however a curved surface or a step are also considered. The sloped surface **124** can be produced by grinding the outer bearing ring **104**. Other fine detail production methods are also possible. The radially outer surface **112** of the inner bearing ring **102** extends at a constant radial distance from a center of the bearing, which is illustrated as a horizontal surface in the cross-section of FIG. 1.

(9) Still with reference to FIG. 1, a sealing arrangement **110** is located within the gap **108**. The sealing arrangement **110** includes at least a first annular body **132**, a first elastomeric lip seal **134** molded onto and partially enveloping the first annular body **132**, a second annular body **136**, and an elastomeric sealing protrusion **122** also molded onto the first annular body **132**. The first annular elastomeric lip seal **134** also contacts the inner bearing ring **102**.

(10) In one embodiment of the sealing arrangement **110**, as shown in FIG. 1, the sealing arrangement **110** includes the first annular body **132** having an axially extending first leg **150**, a radially inwardly directed bend **152** that transitions to a radially extending second leg **154**, and the elastomeric sealing protrusion **122** being located in an area of the radially inwardly directed bend **152**. The first annular body **132** also includes an axially extending third leg **156** and the first elastomeric lip seal **134** is connected to the third leg **156**. A radially outwardly directed bend **158** can extend between the third leg **156** and the radially extending second leg **154** such that the first annular body **132** can be formed in one piece.

(11) The second annular body **136** includes an axially extending fourth leg **138** press fit to the radially outer surface **112** of the inner bearing ring **102**. The axially extending fourth leg **138** transitions into a radially extending fifth leg **140**, and an axially extending sixth leg **142** extends from a radially outer end of the fifth leg **140** to define a generally U-shaped cross-section. The third leg **156** extends into the generally U-shaped cross-section to define a labyrinth seal and a second elastomeric lip seal **160** extends from the first annular body **132** to the second annular body **136**.

(12) The sealing arrangement **110** is also intended to retain and/or include an encoder **126** to deliver a magnetic signal for wheel speed sensor to measure the wheel speed. In one embodiment of the disclosure, the encoder **126** can be located on an axially exterior side **128** of the sealing arrangement **110**, for example on the second annular body **136**, and can be magnetized on an axially exterior side **130** thereof. In this case, the first annular body **132** can be made of magnetic stainless steel or mild steel, such as ferritic steel. The encoder **126** can also be oriented axially and be magnetized radially inward at a location **137** radially outward from the axially extending leg sixth **142**. In this instance the first annular member **132** can be made of a non-magnetic stainless steel such as austenitic steel. At the same time, the outer bearing ring **104** must have at least on one circumferential position an opening that extends in axial direction so that a magnetic field provided by the radially magnetized encoder **126** can pass non-magnetic stainless steel of the first annular

member **132** and be picked up by wheel speed sensor. The first annular body **132** and the encoder **126** define a gap **166** at a radially inner portion of the first annular body **132**.

(13) The elastomeric sealing protrusion **122** can contact the second radial surface portion **118** in the installed position and can optionally also contact the sloped surface **124** extending between the first and second radial surface portions **116**, **118** of the outer bearing ring **104**. The first annular body **132** is configured to be press fit against the first radial surface portion **116** of the outer bearing ring **104** and against sloped surface **124**. A maximum diameter (D1) (shown via dashed lines) of the sealing arrangement **110**, defined by the first leg **150** of the first annular body **132**, is greater than a diameter (D2) (shown via dashed lines) defined by the second radial surface portion **118** of the outer bearing ring **104**.

(14) Referring to FIGS. 2A-2C, a method of inserting the bearing sealing arrangement **110** is shown. FIG. 2A shows the inner bearing ring **102**, the outer bearing ring **104**, and part of the sealing arrangement **110** with the first annular body **132** and the first elastomeric lip seal **134** outside of the bearing assembly. FIG. 2B shows a partially inserted sealing arrangement **110** with the first annular body **132** and the first elastomeric lip seal **134**, and the elastomeric sealing protrusion **122** not contacting the first radial surface portion **116** of the outer bearing ring **104**. FIG. 2C shows a fully inserted sealing arrangement **110**, with the elastomeric sealing protrusion **122** pressed against the second radial surface portion **118**, and the first annular body **132** press fit against the first radial surface portion **116** and against sloped surface **124** for an axial stop, along with the second annular body **136** and the encoder **126** closing in the sealing arrangement from an axially external side.

(15) The systems of the present disclosure, as described above and shown in the drawings, provide for a sealing arrangement with superior properties, allowing for a longer useful life due to both repeatable and precision positioning as well as a decreased likelihood of shearing off a portion of the seal during assembly.

(16) FIG. 3 shows a cross-section of an alternate bearing assembly **200**. The bearing assembly **200** includes an inner bearing ring **202** and an outer bearing ring **204** arranged with rolling bodies **206** therebetween. A gap **208** between the inner bearing ring **202** and the outer bearing ring **204** provides space for a sealing arrangement **210**. The gap **208** is defined by a radially outer surface **212** of the inner bearing ring **202** and a radially inner surface **214** of the outer bearing ring **204**. The radially inner surface **214** of the outer bearing ring **204** can include at least a first radial surface portion **216** and a second radial surface portion **218**. The second radial surface portion **218** is radially inward of the first radial surface portion **216** and the two radial surface portions **216**, **218** are connected by a sloped surface **224**, however a curved surface or a step are also considered. The sloped surface **224** can be produced by grinding the outer bearing ring **204**. Other fine detail production methods are also possible. The radially outer surface **212** of the inner bearing ring **202** extends at a constant radial distance from a center of the bearing, which is illustrated as a horizontal surface in the cross-section of FIG. 3.

(17) Still with reference to FIG. 3, a sealing arrangement **210** is located within the gap **208**. The sealing arrangement includes at least a first annular body **232**, a first elastomeric lip seal **234** molded onto and partially enveloping the first annular body **232**, a second annular body **236**, and an elastomeric sealing protrusion **222** also molded onto the first annular body **232**. The first annular elastomeric lip seal **234** also contacts the axially extending leg fourth **238** of the second annular body **236**.

(18) In one embodiment of the sealing arrangement **210**, as shown in FIG. 3, the sealing arrangement **210** includes the first annular body **232** having an axially extending first leg **250**, a radially inwardly directed connecting leg **252** that transitions to a radially extending second leg **254**, and the elastomeric sealing protrusion **222** is located in an area of the radially inwardly directed connecting leg **252**. The first elastomeric lip seal **234** includes three lips **234a**, **234b**, and **234c**, with the second lip **234b** contacting the second annular body **236**, and the third lip **234c**

contacting or being also being in close proximity to the second annular body **236**.

(19) The second annular body **236** includes an axially extending fourth leg **238** press fit to the radially outer surface **212** of the inner bearing ring **202**. The axially extending fourth leg **238** transitions into a radially extending fifth leg **240**, which contacts the second lip **234b** of the first elastomeric lip seal **234**. The third lip **234c** contacts or is in close proximity to the axially extending fourth leg **238**. The first lip **234a** forms part of a labyrinth seal with a radial outer end **241** of the radially extending fifth leg **240**.

(20) The sealing arrangement **210** is also intended to retain and/or include an encoder **226** to deliver a magnetic signal for wheel speed sensor to measure the wheel speed. In one embodiment of the disclosure, the encoder **226** can be located on an axially exterior side **228** of the sealing arrangement **210**, for example on the second annular body **236**, and can be magnetized on an axially exterior side **230** thereof. In this case, the first annular body **232** can be made of magnetic stainless steel or mild steel, such as ferritic steel. The first annular body **232** and the encoder **226** can define a gap **266** at a radially inner portion of the first annular body **232**.

(21) The elastomeric sealing protrusion **222** can contact the second radial surface portion **218** in the installed position and can optionally also contact the sloped surface **224** extending between the first and second radial surface portions **216**, **218** of the outer bearing ring **204**. The first annular body **232** is configured to be press fit against the first radial surface portion **216** of the outer bearing ring **204** and against sloped surface **224**. The maximum diameter (D1) (shown via dashed lines) of the sealing arrangement **210**, defined by the first leg **250** of the first annular body **232**, is greater than the diameter (D2) (shown via dashed lines) defined by the second radial surface portion **218** of the outer bearing ring **204**.

(22) FIG. 4 shows a cross-section of an alternate sealing arrangement **310** of bearing assembly **300**. The bearing assembly **300** includes an inner bearing ring **302** and an outer bearing ring **304** arranged with rolling bodies **306** therebetween. A gap **308** between the inner bearing ring **302** and the outer bearing ring **304** provides space for a sealing arrangement **310**. The gap **308** is defined by a radially outer surface **312** of the inner bearing ring **302** and a radially inner surface **314** of the outer bearing ring **304**. The radially inner surface **314** of the outer bearing ring **304** can include at least a first radial surface portion **316** and a second radial surface portion **318**. The second radial surface portion **318** is radially inward of the first radial surface portion **316** and the two radial surface portions **316**, **318** are connected by a sloped surface **324**, however a curved surface or a step are also considered. The sloped surface **324** can be produced by grinding the outer bearing ring **304**. Other fine detail production methods are also possible. The radially outer surface **312** of the inner bearing ring **302** extends at a constant radial distance from a center of the bearing, which is illustrated as a surface parallel to the rotational axis in the cross-section of FIG. 4.

(23) Still with reference to FIG. 4, the sealing arrangement **310** is located within the gap **308**. The sealing arrangement **310** includes at least a first annular body **332**, a first elastomeric lip seal **334** molded onto the first annular body **332**, and a second annular body **336**. The first annular body **332** includes an axially extending first leg **350**, a radially inwardly directed connecting leg **352** that includes an axially outward offset section **352a**. The first elastomeric lip seal **334** includes two lips **334a** and **334b** with the second lip **334b** contacting the second annular body **336**. The first elastomeric lip seal **334** is also molded and covers an outer surface **333** of the first annular body **332** and an axially distal end **351** of the first annular body **332**, and wraps onto a portion of the radially outer surface **335** of the first annular body **332**. In this configuration, the first elastomeric lip seal **334** provides a sealing interface with the outer bearing ring **304** instead of the sealing arrangement **310** including a separate molded sealing component, such as elastomeric sealing protrusion **122**, **222**.

(24) Having thus described the exemplary embodiments in detail, it is to be appreciated and will be apparent to those skilled in the art that many physical changes, only a few of which are exemplified in the detailed description, could be made without altering the inventive concepts and principles

embodied therein. It is also to be appreciated that numerous embodiments incorporating only part of the disclosed embodiment are possible which do not alter, with respect to those parts, the inventive concepts and principles embodied therein. The present embodiments and optional configurations are therefore to be considered in all respects as exemplary and/or illustrative and not restrictive, the scope that is indicated by the appended claims rather than by the foregoing description, and all alternate embodiments and changes to this embodiment which come within the meaning and range of equivalency of said claims are therefore to be embraced therein.

LOG OF REFERENCE NUMERALS

(25) assembly **100, 200, 300** inner bearing ring **102, 202, 302** outer bearing ring **104, 204, 304** rolling bodies **106, 206, 306** gap **108, 208, 308** sealing arrangement **110, 210, 310** radially outer surface **112, 212, 312** radially inner surface **114, 214, 314** first radial surface portion **116, 216, 316** second radial surface portion **118, 218, 318** elastomeric sealing protrusion **122, 222** sloped surface **124, 224, 324** encoder **126, 226** axially exterior side **128, 228** axially interior side **130, 230** first annular body **132, 232, 332** axially exterior surface of first annular body **333** first elastomeric lip seal **134, 234, 334** first lip **234a, 334a** second lip **234b, 334b** third lip **234c** radially exterior surface of first annular body **335** second annular body **136, 236, 336** fourth leg **138, 238** fifth leg **140, 240** sixth leg **142** radial outer end of the fifth radially extending leg **241** first leg **150, 250, 350** distal end of the first leg **351** radially inward bend **152** radially inward connecting leg **252, 352** axially outward offset section of radial leg **352a** second leg **154, 254** third leg **156** radially outward bend **158** second elastomeric lip seal **160** gap **166, 266**

Claims

1. A bearing assembly comprising: an inner bearing ring; an outer bearing ring; rolling bodies located between the inner bearing ring and the outer bearing ring; a gap defined by a radially outer surface of the inner bearing ring and a radially inner surface of the outer bearing ring, the radially inner surface of the outer bearing ring having a first radial surface portion that is offset a greater distance from a center of the bearing assembly than a second radial surface portion, the first and second radial surface portions being connected by a sloped surface; and a sealing arrangement located in the gap, the sealing arrangement including: a metallic first annular body configured to contact the sloped surface of the outer bearing ring, including an axially extending first leg being configured for a press fit with the first radial surface portion in an installation position; wherein the axially extending first leg includes a radially inwardly directed bend that transitions to a radially extending second leg, and, an elastomeric sealing protrusion on the first annular body adjacent to the sloped surface at the radially inwardly directed bend; and wherein the first annular body further comprises an axially extending third leg extending from a radially inner end of the radially extending second leg to define a U-shaped cross-section, and a first elastomeric lip seal; and a second annular body including an encoder and having an axially extending fourth leg, wherein the fourth axially extending leg transitions into a radially extending fifth leg, and an axially extending sixth leg extending from a radially outer end of the radially extending fifth leg to define a U-shaped cross-section.
2. The bearing assembly of claim 1, wherein the axially extending third leg includes a radially outwardly directed bend.
3. The bearing assembly of claim 1, wherein a second elastomeric lip seal extends from the first annular body to the second annular body.
4. The bearing assembly of claim 1, wherein the elastomeric sealing protrusion includes a maximal outer diameter greater than a diameter defined by the second radial surface portion of the outer bearing ring.
5. The bearing assembly of claim 4, wherein the elastomeric sealing protrusion includes a maximal outer diameter smaller than an outer diameter of the first leg.

6. The bearing assembly of claim 1, wherein the first annular body is metallic.
 7. The bearing assembly of claim 1, wherein the first annular body is non-magnetic.
 8. The bearing assembly of claim 1, wherein the encoder is located on an axially exterior side of second annular body.
 9. The bearing assembly of claim 8, wherein the encoder is magnetized on an axially exterior side thereof or on a radially inward side thereof.
 10. The bearing assembly of claim 8, wherein the first annular body and the encoder define at least one gap therebetween.
 11. The bearing assembly of claim 1, wherein the axially extending third leg extends into the U-shaped cross-section of the second annular body to define a labyrinth seal.
 12. The bearing assembly of claim 1, wherein the first leg, the second leg, and the third leg comprise a unitary, monolithic metal piece.
 13. The bearing assembly of claim 1, wherein the fourth leg, the fifth leg, and the sixth leg comprise a unitary, monolithic metal piece.
 14. A bearing assembly comprising: an inner bearing ring; an outer bearing ring; rolling bodies located between the inner bearing ring and the outer bearing ring; a gap defined by a radially outer surface of the inner bearing ring and a radially inner surface of the outer bearing ring, the radially inner surface of the outer bearing ring having a first radial surface portion that is offset a greater distance from a center of the bearing assembly than a second radial surface portion, the first and second radial surface portions being connected by a sloped surface; and a sealing arrangement located in the gap, the sealing arrangement including: a first metallic annular body configured to contact the sloped surface of the outer bearing ring, including an axially extending first leg being configured for a press fit with the first radial surface portion in an installation position, the axially extending first leg includes a radially inwardly directed bend that transitions to a radially extending second leg; wherein the first annular body further comprises an axially extending third leg, extending from a radially inner end of the radially extending second leg to define a U-shaped cross-section, and a first contacting elastomeric lip seal; and an elastomeric sealing protrusion on the first annular body adjacent to the sloped surface at the radially inwardly directed bend, the elastomeric sealing protrusion contacting the second radial surface portion.
 15. The bearing assembly of claim 14, wherein the axially extending third leg includes a radially outwardly directed bend.
 16. The bearing assembly of claim 15, wherein the axially extending third leg extends into the U-shaped cross-section to define a labyrinth seal.
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